DNA Extractions from 500 mL culture for long read genomics

This is a standard Phenol-Chloroform Sucrose Lysis Buffer extraction protocol done in either multiple 2 mL tubes or a single 50mL centrifuge tube.
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For up to 500 ml culture, pellet in the 500 ml Oak Ridge tubes (they only hold ~450 ml, so best to either split over 2 Oak Ridge tubes, or pellet 250 ml twice). Once cells are in a pellet, resuspend in 10 ml of sterile artificial seawater, or 3X PBS, transfer to a 50 ml falcon tubes. Pellet again, remove supernatant, and store at -80°C until needed.

Buffers and reagents needed

Lysis Buffer (make up the total volume with Nanopure water and filter sterilize)

Final Concentration	Stocks .	500 mL	250 mL	50 mL
40 mM EDTA	0.5 M EDTA (pH 8)	40 mL	20 mL	4 mL
50 mM Tris-Cl	1 M Tris-Cl (pH 8)	25mL	12.5 mL	2.5 mL
0.75 M Sucrose	Sucrose	128 g	64 g	12.8 g
0.2 M NaCl	NaCl	23 g	11.5 g	2.3 g

Lysozyme stock is made to 50 mg/mL in Lysis Buffer (pH 8) for final concentration of ~1 mg/mL in the reaction; make fresh before use; do not save; Lysozyme has optimal activity at pH 8.

RNAse A (optional), 4-5U per ug of DNA, stock is 10mg/ml of ~5000U/mg. This will likely only cut single stranded RNA since the salt concentration is high. At a lower salt concentration, this will cleave double-stranded RNA and DNA-RNA hybrids.

20% SDS solution (final 1% SDS; Pro K is highly active at 0.5-1% SDS)

Proteinase K solution (~600U activity/mL, ~15U activity needed per reaction)

Chloroform : Isoamyl alcohol (IAA), 24:1

Phenol, Tris buffer equilibrated to pH 8.0. Phenol is in the lower organic phase unless the aqueous phase is dense due to >0.5M NaCl or >10% sucrose. You can test which phase is aqueous by adding a drop of your lysis buffer to the separated phenol/aqueous mixture to see where it goes. Phenol should be clear and has gone bad if it turns yellow or pink unless 0.1% hydroxyquinoline (0.1 g per 100 mL phenol phase) was added to prevent oxidation which will make it yellow. Phenol stored at 4°C should last ~3 months without and ~12 months with hydroxyquinoline.

DNA Extraction Protocol using 50 mL tubes

Supplies:

50 mL Oakridge tubes (Nalgene cat# 3119-0050) actually hold 40 mL and are rated for 50,000 g.
50 mL Falcon Conical Centrifuge tube, rated for 16,000 g.
50 mL tube heat block (set at 300 RPM to gently shake samples)
Centrifuge with rotor for Oakridge tubes

Directions:

1. Add 20 mL Lysis Buffer to the tube with the cell pellet and gently resuspend cell pellet.

2. Add 400 uL Lysozyme (50mg/mL solution).

3. Incubate at 37°C for 30 minutes.

4. Add 1 mL 20% SDS and gently invert a few times to incorporate.

5. Then add 500 uL Proteinase K solution and gently invert a few times to incorporate.

6. Incubate at 55°C for 30 minutes.
    a. *Optional: Add 80 ug/ml of RNAse A and incubate at 37°C for 30 minutes*

7. Transfer 20 mL and no more of the lysates to a 50 mL Polypropylene Oakridge tube (do not use polycarbonate with phenol).

8. Add equal volume, 20 mL Phenol : Chloroform : Isoamyl alcohol (IAA) (25:24:1, pH 8.0). Shake the tube a few times to mix. Spin 5 minutes at 13,000 g.

9. Use a 10 mL pipet and transfer the aqueous phase to a new 50 mL Oakridge tube.
10. Add equal volume (~10 mL) Chloroform:IAA (24:1). Shake a few times to mix. Spin 5 minutes at 13,000 g.

11. Remove the aqueous phase to a clean tube and use either an Centricon/Amicon OR an Ethanol precipitation below to remove the salts.

DNA Extraction Protocol using 2 mL tubes

Supplies:

2 mL Eppendorf Safe-Lock Tubes (Forensic gr, 50 tubes/pk, cat# 0030123620, rated for 25,000 g)
50 mL tube heat block (set at 300 RPM to gently shake samples)
Microcentrifuge

Directions:

1. Gently resuspend cell pellet in 4 mL of Lysis Buffer.

2. Add 80 uL Lysozyme (50 mg/mL solution).

3. Incubate at 37°C for 30 minutes.

4. Add 200 uL 20% SDS and gently invert a few times to incorporate.

5. Then add 100 uL Proteinase K solution and gently invert a few times to incorporate.

6. Incubate at 55°C for 30 minutes.
    a. *Optional: Add 80 ug/ml of RNAse A and incubate at 37°C for 30 minutes*

7. Transfer lysates (~900 uL) to 4 – 6, 2 mL Eppendorf Safe-Lock tube.

8. Add equal volume, (~900 uL) Phenol : Chloroform : Isoamyl alcohol (IAA) (25:24:1, pH 8.0). Shake the tube a few times to emulsify. Spin 5 minutes at 16,000-20,000 g.

9. Use a 1 mL pipet and transfer the aqueous phase to a new 2 mL tube. *(Note: Make a large bore tip by cutting off tip of pipet straight across with flame sterile scissors to help remove the aqueous phase without disturbing the protein layer)*

10. Add equal volume (~900 uL) Chloroform:IAA (24:1). Shake for a few seconds to emulsify. Spin 5 minutes at 16,000-20,000 g.

11. Remove the aqueous phase (~700 uL) with a large bore tip to a clean 2 mL tube and use either an Vivaspin/Amicon OR an Ethanol precipitation below to remove the salts.

Ethanol Precipitation (Molecular Cloning, Sambrook et al, A8.12)

Use the precipitation described below for a 2 mL tube or a 50 mL Oakridge tube. This precipitation takes advantage of the Lysis Buffer above, which already has 0.2 M NaCl and is an exceptionally good precipitation salt for SDS based buffers at this molarity.

50 mL Tube

10 mL DNA solution
22 mL ice cold 100% Ethanol, molecular biology grade (69% EtOH)

2 mL Tube

600 uL DNA solution
1320 uL ice cold 100% Ethanol, molecular biology grade (69% EtOH)

A. Invert the tube 3-6 times to mix and precipitate DNA overnight at -20°C. If in a hurry precipitate for at least 1 hour on ice but yields might be lower.

B. Spin for 1 hour at 4°C at 13,000 to 20,000 g.

C. Remove supernatant while carefully avoiding the pellet that will extend up the side of the tube.

D. Optional salt rinse: Gently add 1 mL (2mL tubes) or 37 mL (50 mL tube) ice cold 70% Ethanol down side of tube and spin 15 minutes at 4°C at 13,000 to 20,000 g. Keep the same orientation for each tube as previous spin.

E. Blot the open tube up-side-down onto a clean kimwipe to remove most ethanol and dry using one of these methods. If not completely dry PCR can be inhibited, but don't over-dry or pellet may not resuspend.
    a. Place open tube in a 45°C 50 mL heat block for 15 minutes or until dry.
    b. Leave open in a clean laminar flow hood overnight at room temperature.

F. Resuspend DNA pellet in 50 uL to 1 mL 10 mM Tris (pH8) or TE buffer (10mM Tris, 1mM EDTA, pH8). Make sure to pipet up and down along side of the tube where the pellet should be. Vortex 10 sec and let sit at room temperature for 5 minutes and vortex again for a few seconds.

G. DNA is ready to use or store at -20 to -80°C. For long read sequencing of genomic DNA avoid freeze thawing – ideally measure concentration after spinning via qubit (quantity) and/or nanodrop (purity), and run on a DNA gel with 24 kb ladder to ensure minimal degradation. If you don't do it the day of, store at 4°C overnight and measure the next day.

Vivaspin/Amicon 100 kDa cut-off Column Cleanup

1. Transfer upper aqueous phase ~700 uL (can pool aqueous phase from all 2 mL tubes that came from the same pellet) or ~10 mL (50 mL tube) to a Vivaspin/Amicon 100 kDa cut-off column in a 15mL or 50mL tube.

2. Spin at 1000 g for 20 min or longer as needed until 50 uL or 1 mL left in column. (max speed is 4000 g; Make sure orientation of the column is correct! The column membrane panels should be parallel to the top of the centrifuge)
3. Wash DNA by adding 1 mL or 10 mL 10 mM Tris (pH8) to the column. *(Note: TE buffer (10 mM Tris, 1 mM EDTA, pH 8) gives more stability to the DNA and can be used instead of Tris as long as the downstream protocol is not sensitive to EDTA, which can slightly inhibit PCR)*
4. Spin at 1000 g until 50 uL or 1 mL is left in the column.
5. Repeat wash steps 3) & 4) two more times.
6. Remove the 50 uL or resuspend in desired volume and transfer from the column to a clean microfuge tube. (*Note: You can add another equal volume to the column and pipette over the membrane a few times after removing eluate to collect any remaining DNA.)*
    a. *If you need even more concentrated, you can transfer the first and second eluate to a 1 ml Vivaspin 100kDa column and spin at 6000 x g for 10 mins. If DNA was not clean enough, in this step you can wash DNA again in same buffer you were using to resuspend DNA.*
7. Store at -20 to -80°C. For long read sequencing of genomic DNA avoid freeze thawing – ideally measure concentration after spinning via qubit (quantity) and/or nanodrop (purity), and run on a DNA gel with 24 kb ladder to ensure minimal degradation. If you don't do it the day of, store at 4°C overnight and measure the next day.
    a. *Depending on the volumes you need, you can aliquot ~6 ul of DNA and save at 4°C, and store the remainder at –20 to -80°C. This should be enough for qubit (2 ul), nanodrop (2 ul), and a DNA gel (likely you will dilute 1:10 or 1:100)*